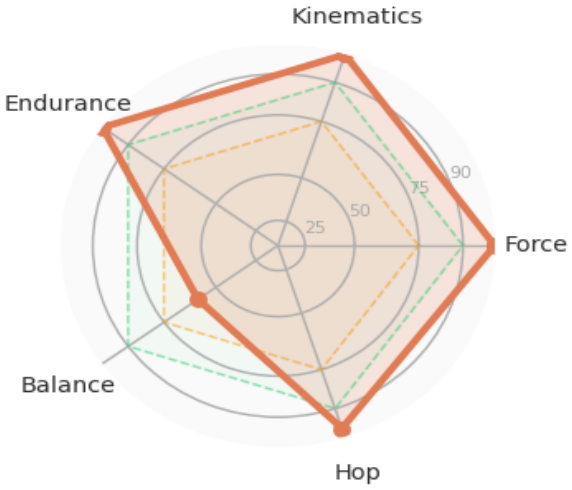
BYoung Physical Therapy

Patient: Jonathan Demay • **Date:** July 20, 2026
Surgical Side: Left • **Triple Hop:** Surg 22'0" / NS 22'8" / LSI 97.1%

Return to Sport Assessment

82 **RTS Score Progressing**
1 scored red flag

Return to Sport Profile



▲ STRENGTHS

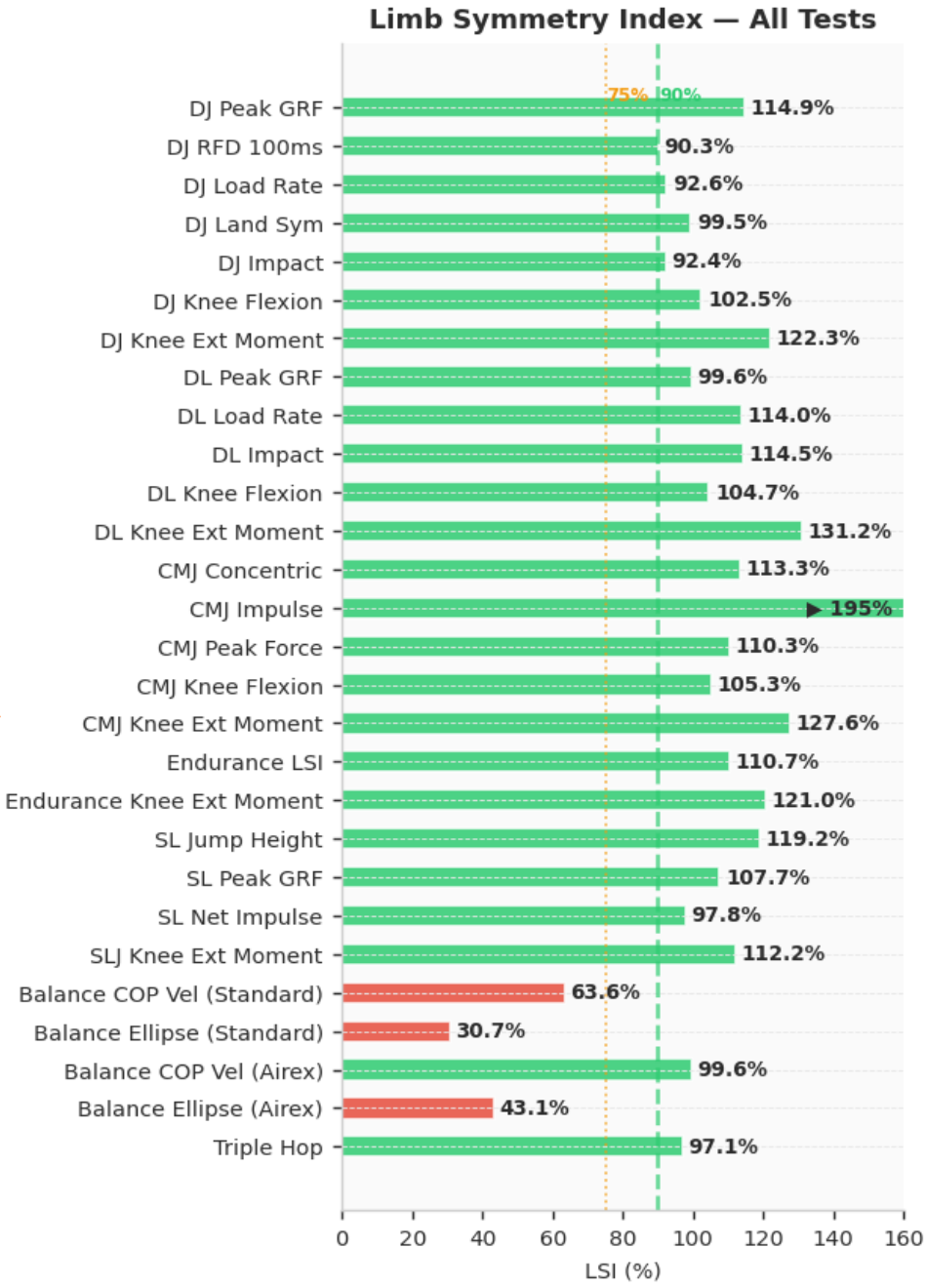
- Single-leg jump (100/100)
- DJ Peak GRF sym. (100/100)
- Endurance symmetry (100/100)
- DL Load rate sym. (100/100)

▼ DETRACTORS

- RSI (reactive strength) (31/100)
- Fatigue drift (62/100)

Performance 86/100 → ×0.95 red-flag penalty (1 scored) → = 82/100

Limb Symmetry Index — All Tests



CLINICAL FLAGS

- Balance Ellipse (Standard): LSI 30.7% — significant asymmetry
- Balance Ellipse (Airex): LSI 43.1% — significant asymmetry
- DJ Knee Valgus: Δ +7.2° — exceeds 6° threshold
- DL Knee Valgus: Δ +8.4° — exceeds 6° threshold
- CMJ Knee Valgus: Δ +8.3° — exceeds 6° threshold
- Balance COP Vel (Standard): LSI 63.6% — significant asymmetry

● ≥90% Symmetric ● ≥75% Progressing ● <75% Deficit LSI = Surgical ÷ Non-Surgical × 100 | Δ = Surgical – Non-Surgical (absolute difference; used for small-angle kinematics)

DROP JUMP

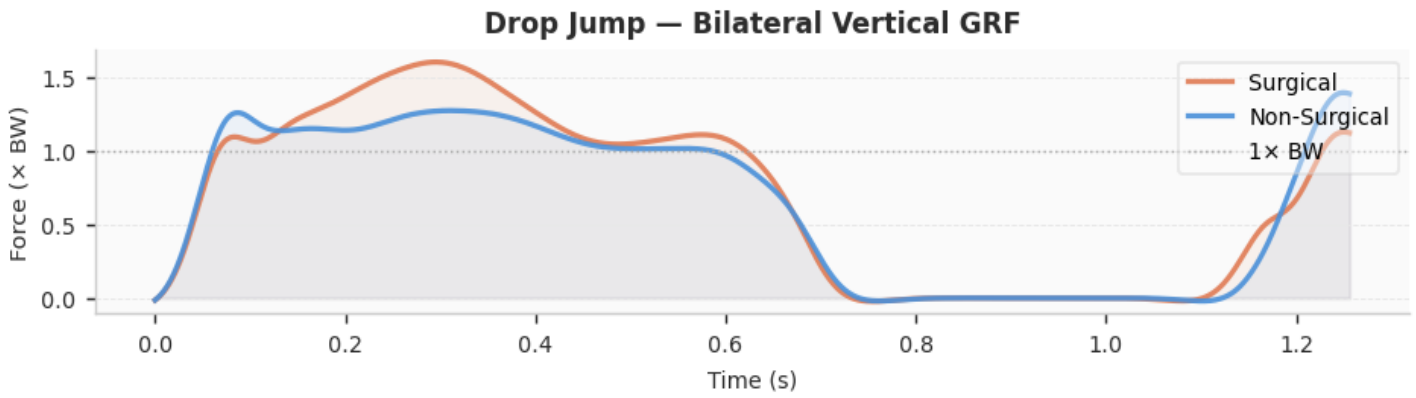
Metric	Surgical	Non-Surg	LSI / Δ
Peak GRF (N)	1359.9	1183.6	114.9%
Impulse (N·s)	733.7	683.0	107.4%
RFD 100ms (N/s)	11194.8	12404.2	90.3%
Loading Rate 50ms (N/s)	13434.0	14506.7	92.6%
Landing Sym 200ms (%) *(bilateral)*	—	—	99.5%
Impact Transient (N)	683.2	739.4	92.4%
Time-to-Peak (ms)	295.0	1249.0	—
Jump Height (in)	14.8	—	—
Contact Time (s)	0.710	—	—
RSI (m/s)	0.53	—	—
Peak Knee Flexion (°)	109.2	106.6	102.5%
Peak Knee Valgus (°)	9.2	2.0	Δ +7.2°
Peak Tibial IR (°)	-7.4	-1.7	Δ -5.7°
Tibial Rot Excursion (°)	6.1	7.8	Δ -1.6°
Peak Hip Adduction (°)	0.4	-1.6	Δ +2.0°
Peak Knee Ext Moment (Nm)	245.4	200.6	122.3%
AP COP Excursion (mm)	145.0	141.1	—

CLINICAL INTERPRETATION

RSI 0.53 m/s with 0.71 s contact (⚠ below RTS target ≥ 1.30). Stiff-landing pattern likely — progressive plyometric loading indicated.

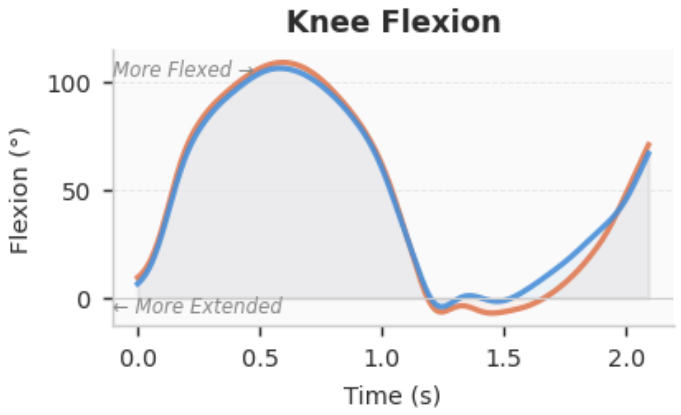
Landing symmetry 99% LSI: Good bilateral loading at contact.

Targets: RSI ≥ 1.30 m/s · Landing LSI 90–110% · Load rate LSI ≥ 90% · Knee flexion ≥ 60° · Valgus < 8° · Tibial IR < 10°.



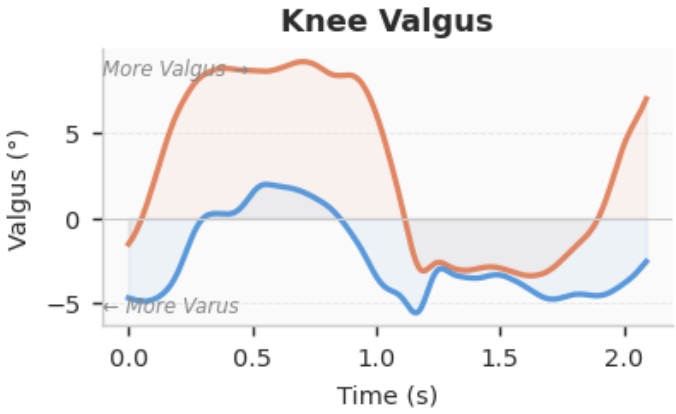
Bilateral GRF (normalised to BW). Orange = surgical, Blue = non-surgical.

Landing symmetry 99% at 200ms — symmetric bilateral loading.

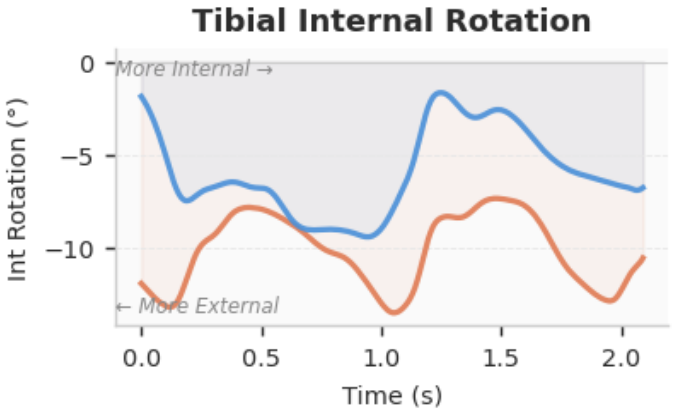


Knee flexion. Target ≥ 60° at contact. Reduced depth = stiff landing strategy.

Surgical limb knee flexion depth is symmetric with the non-surgical side (LSI 103%) — meets return-to-sport threshold.



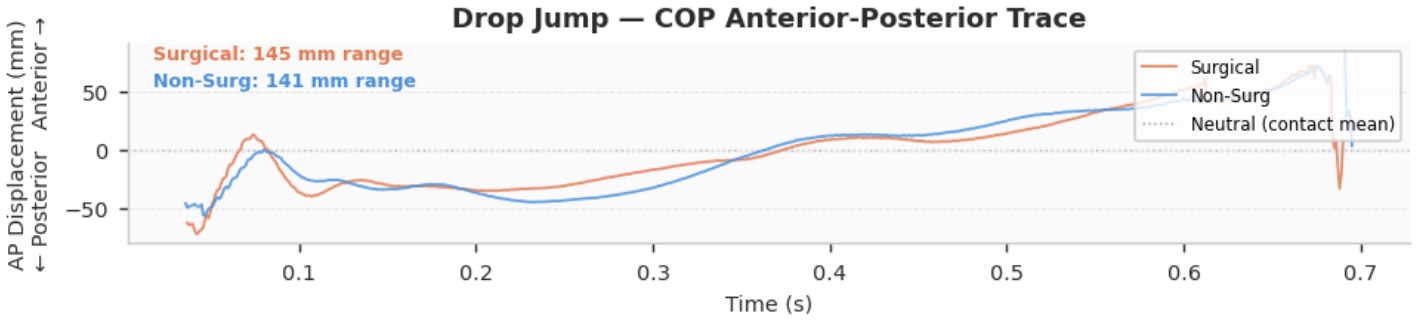
Knee valgus. Target < 8°. Elevated valgus = medial collapse / ACL re-injury risk.



Tibial internal rotation. IR > 10° at landing = increased ACL rotational load.

***Left panel:** Surgical knee valgus: 9.2° vs non-surgical 2.0° (Δ +7.2°) — side-to-side difference exceeds 6°; targeted intervention recommended.*

***Right panel:** Surgical tibial IR: -7.4° vs non-surgical -1.7° (Δ -5.7°) — mild side-to-side asymmetry; monitor closely.*



AP COP displacement during the first contact (zero-centred). Smaller excursion on surgical limb = restricted fore-aft weight transfer / guarded landing strategy.

AP COP excursion: surgical 145 mm vs non-surgical 141 mm — symmetric fore-aft weight shift.

Surgical limb bilateral landing symmetry is symmetric with the non-surgical side (LSI 99%) — meets return-to-sport threshold.

ABOUT THIS TEST

The Drop Jump is a reactive bilateral landing task performed from a standard box height. The athlete drops off the box, absorbs the impact with both limbs simultaneously, then immediately jumps for maximum height. It tests the athlete's ability to rapidly store and release elastic energy (reactive strength) and to absorb high-impact loads symmetrically. Key concerns after ACL reconstruction include stiff-knee landing (reduced flexion, elevated impact transient), medial knee collapse (valgus), and side-to-side force asymmetries that indicate ongoing limb offloading. The Reactive Strength Index (RSI = jump height ÷ contact time) summarises the explosive reactive capacity of the neuromuscular system.

DROP LANDING

Metric	Surgical	Non-Surg	LSI / Δ
Peak GRF (N)	1093.6	1098.2	99.6%
Impulse (N·s)	782.1	656.3	119.2%
Loading Rate 50ms (N/s)	14122.0	12387.3	114.0%
Impact Transient (N)	757.1	661.0	114.5%
Peak Knee Flexion (°)	112.7	107.7	104.7%
Peak Knee Valgus (°)	9.8	1.4	Δ +8.4°
Peak Tibial IR (°)	-7.4	-1.3	Δ -6.1°
Tibial Rot Excursion (°)	5.7	8.2	Δ -2.5°
Peak Hip Adduction (°)	-5.5	-4.7	Δ -0.8°
Peak Knee Ext Moment (Nm)	198.7	151.4	131.2%
AP COP Excursion (mm)	104.9	88.2	—

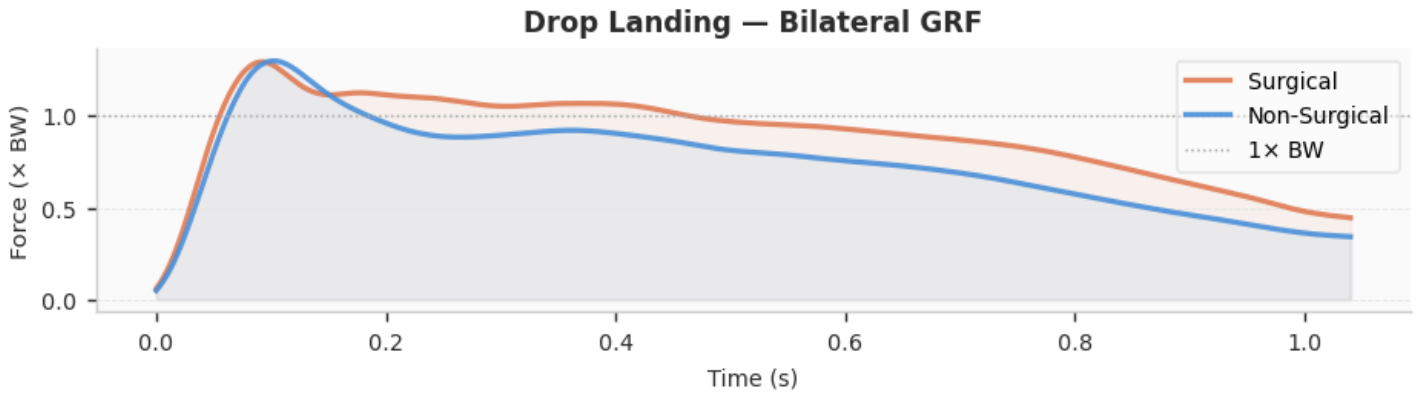
CLINICAL INTERPRETATION

Absorptive landing — assesses shock attenuation without a re-jump. Lower, more gradual force = better control.

Load rate LSI 114%: Surgical limb loading rate 14% higher — NS side protective offloading.

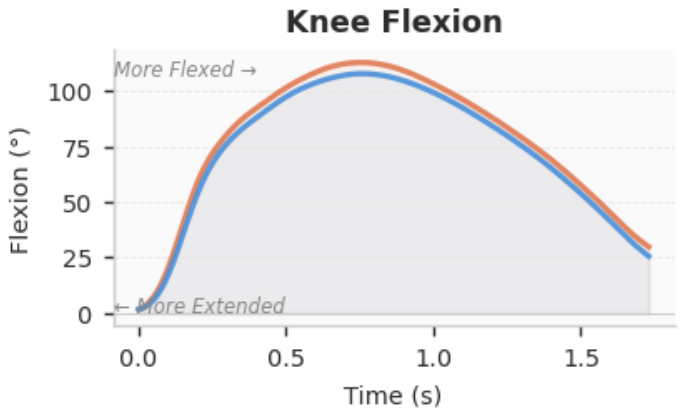
Impact transient LSI 115% — surgical limb initial impact 15% higher.

Targets: Peak GRF < 3× BW · Load rate LSI 90–110% · Impact transient LSI 90–110% · Knee flexion ≥ 60° · Valgus < 8° · Tibial IR < 10°.



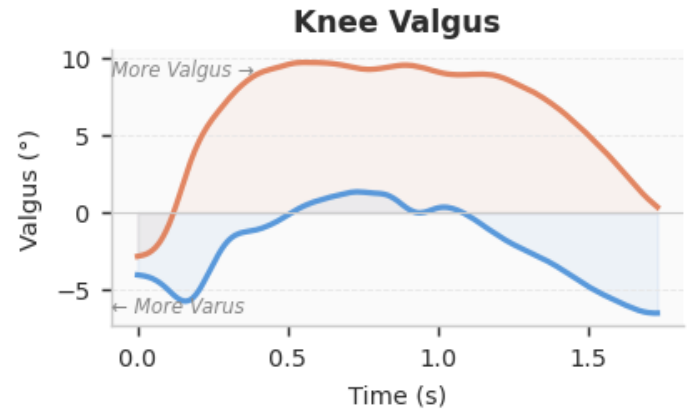
Bilateral GRF. Smooth progressive rise = good shock absorption. Sharp impact spike = stiff landing.

Peak GRF symmetry 100% — meets return-to-sport threshold.

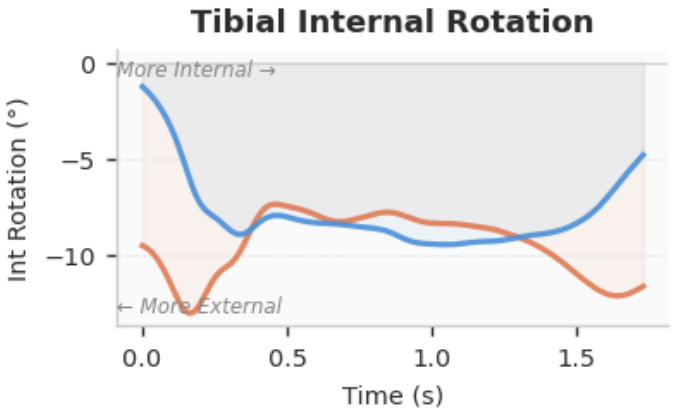


Knee flexion at landing. Reduced depth = stiff strategy / quad inhibition.

Surgical limb landing knee flexion depth is symmetric with the non-surgical side (LSI 105%) — meets return-to-sport threshold.



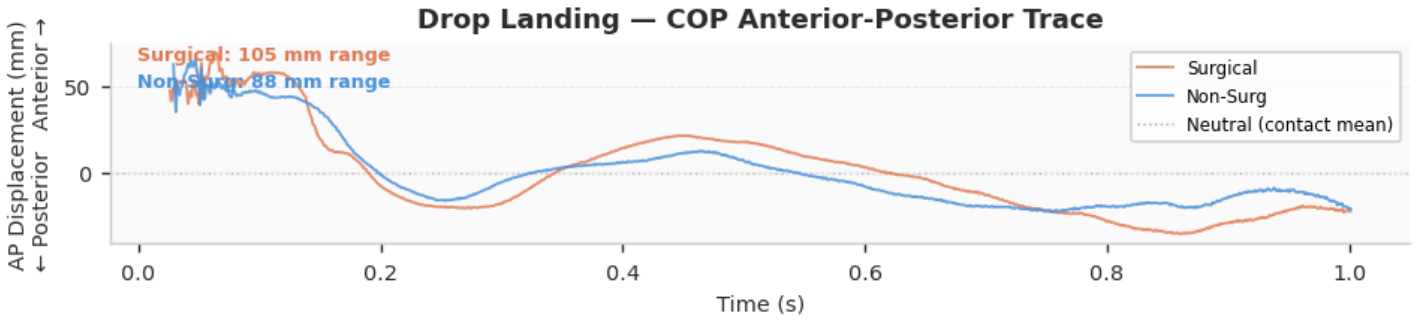
Knee valgus at landing. Target < 8°; higher surgical valgus = elevated ACL stress.



Tibial IR at landing. Combined with valgus = high ACL injury-risk pattern.

Left panel: *Surgical knee valgus: 9.8° vs non-surgical 1.4° (Δ +8.4°) — side-to-side difference exceeds 6°; targeted intervention recommended.*

Right panel: *Surgical tibial IR: -7.4° vs non-surgical -1.3° (Δ -6.1°) — mild side-to-side asymmetry; monitor closely.*



AP COP during landing contact (zero-centred). Posterior bias on surgical side = heel-dominant / reduced forefoot loading.

AP COP excursion: surgical 105 mm vs non-surgical 88 mm — more fore-aft travel on the surgical side — less controlled foot loading.

Surgical limb peak landing force absorption is symmetric with the non-surgical side (LSI 100%) — meets return-to-sport threshold.

ABOUT THIS TEST

The Drop Landing isolates the deceleration phase of a jump: the athlete drops from a box and sticks the landing bilaterally without a subsequent jump. This removes the propulsive demand and focuses entirely on the athlete's ability to safely absorb and dissipate ground reaction forces. After ACL reconstruction, common compensations include a stiff-knee strategy (reduced flexion, sharp impact spike), excessive valgus collapse, and asymmetric weight acceptance—all of which increase mechanical stress at the reconstructed ligament. A smooth, progressive force curve with symmetric knee flexion ≥ 60° and valgus < 8° is the clinical target. Loading rate and impact transient LSI reflect side-to-side shock absorption capacity.

MAX VERTICAL JUMP (CMJ)

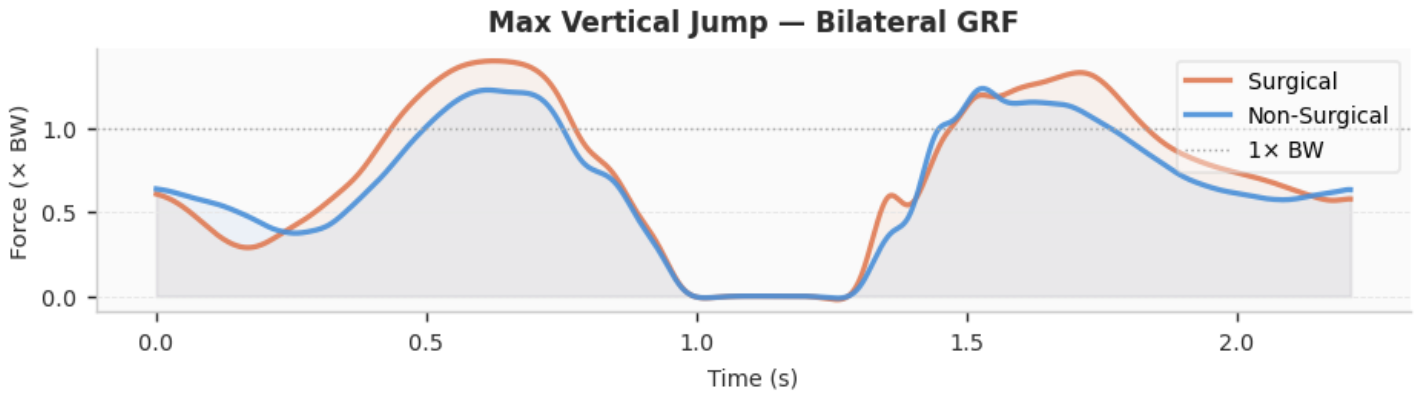
Metric	Surgical	Non-Surg	LSI / Δ
Peak Concentric GRF (N)	1186.4	1047.3	113.3%
Propulsion Impulse (N·s)	949.1	485.8	195.4%
Peak Force Surg / NS (N)	748.8	679.1	110.3%
Propulsion LSI (%) *(bilateral)*	—	—	195.4%
Jump Height (in)	16.0	—	—
Flight Time (s)	0.300	—	—
Unweighting Impulse Surg/NS	185.6	185.2	—
Peak Knee Flexion (°)	123.3	117.0	105.3%
Peak Knee Valgus (°)	10.3	2.0	Δ +8.3°
Peak Tibial IR (°)	-4.0	-1.0	Δ -3.0°
Tibial Rot Excursion (°)	7.8	6.8	Δ +1.0°
Peak Hip Adduction (°)	-1.1	-2.6	Δ +1.5°
Peak Knee Ext Moment (Nm)	236.1	185.1	127.6%
AP COP Excursion (mm)	166.0	155.3	—

CLINICAL INTERPRETATION

Bilateral CMJ — sensitive to subtle propulsion deficits masked at lower intensities.

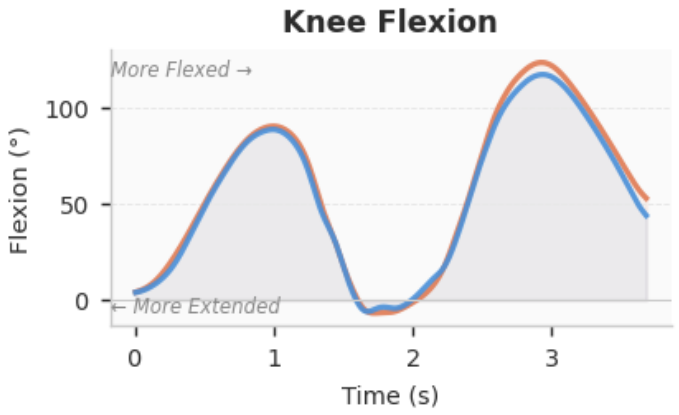
Peak force LSI 110%: Surgical limb generating 10% more force — possible NS compensation.

Propulsion LSI 195%. Targets: ≥ 90% propulsion, ≥ 90% peak force, matched unweighting impulse.



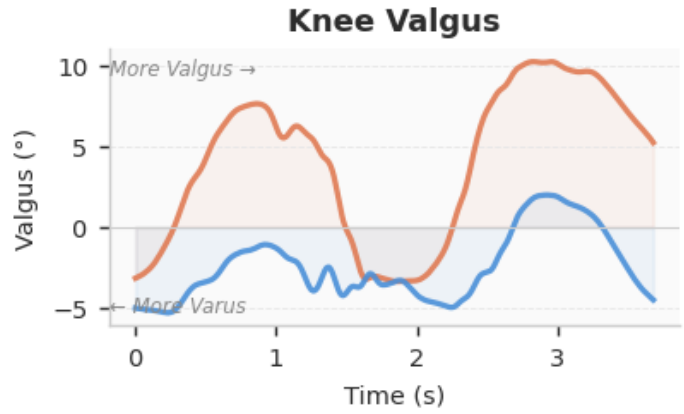
Full CMJ: unweighting → eccentric load → propulsion → flight. Propulsive peaks should be symmetric.

Surgical limb propulsive impulse is symmetric with the non-surgical side (LSI 195%) — meets return-to-sport threshold.

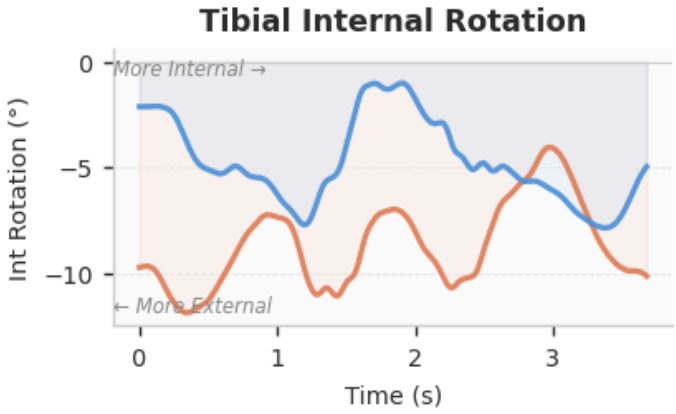


Knee flexion depth. Asymmetric countermovement depth = compensatory loading strategy.

Surgical limb countermovement knee flexion depth is symmetric with the non-surgical side (LSI 105%) — meets return-to-sport threshold.



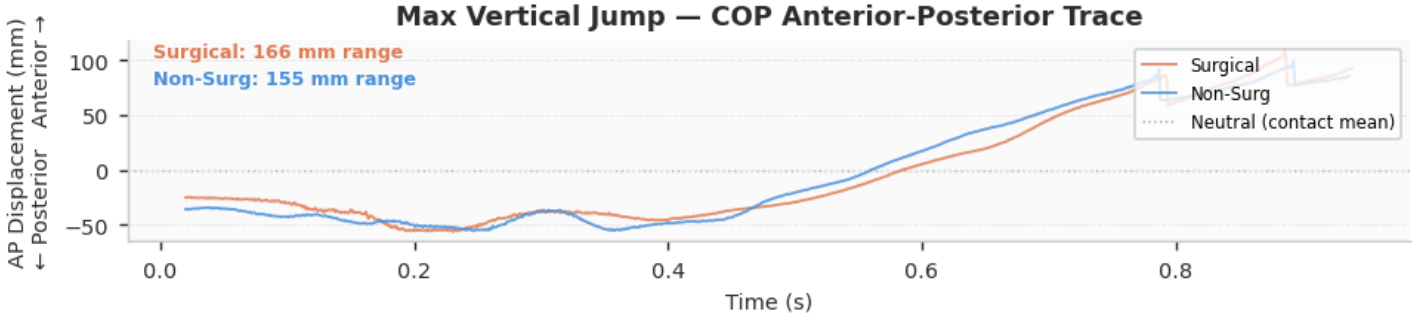
Knee valgus during propulsion. Dynamic valgus under load = ACL stress.



Tibial IR during propulsion. Coupled with valgus = high re-injury risk pattern.

***Left panel:** Surgical knee valgus: 10.3° vs non-surgical 2.0° (Δ +8.3°) — side-to-side difference exceeds 6°; targeted intervention recommended.*

***Right panel:** Surgical tibial IR: -4.0° vs non-surgical -1.0° (Δ -3.0°) — side-to-side difference within acceptable range.*



AP COP during the propulsive contact (zero-centred). Anterior shift during propulsion drives jump height — asymmetric or reduced excursion on surgical side = limb offloading.

AP COP excursion: surgical 166 mm vs non-surgical 155 mm — symmetric fore-aft weight shift.

Surgical limb peak propulsive force is symmetric with the non-surgical side (LSI 110%) — meets return-to-sport threshold.

ABOUT THIS TEST

The Countermovement Jump (CMJ) is the gold-standard bilateral power test. The athlete begins in quiet stance, rapidly descends into a squat (countermovement), then drives upward for maximum jump height. The full GRF waveform reveals each phase: unweighting, eccentric loading, propulsion, and flight. After ACL reconstruction, typical deficits include an asymmetric countermovement depth (surgical limb offloading during the eccentric phase), reduced propulsive impulse, and side-to-side jump height asymmetry. Propulsive LSI < 90% is a common criterion for RTS decisions. Kinematic analysis identifies compensatory strategies—valgus, reduced knee flexion, and tibial rotation—that indicate the surgical limb is not being loaded with full confidence or mechanical efficiency.

ENDURANCE SQUAT (30s)

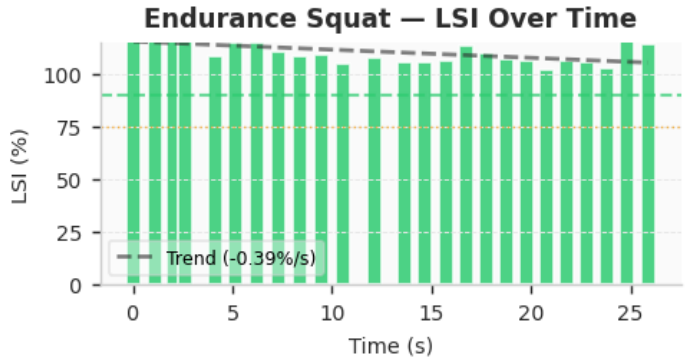
Metric	Surgical	Non-Surg	LSI / Δ
Mean Peak Force LSI (%) *(bilateral)*	—	—	110.7%
LSI First 10s (%) *(bilateral)*	—	—	116.2%
LSI Last 10s (%) *(bilateral)*	—	—	107.9%
Fatigue Drift (Δ%)	-8.3	—	—
Total Reps Detected	25.0	—	—
Mean Peak Force Surg (N)	1011.4	—	—
Mean Peak Force NS (N)	915.1	—	—
AP COP Excursion — Surg / NS (mm)	179.8	165.3	—
Peak Knee Ext Moment (Nm)	222.9	184.2	121.0%

CLINICAL INTERPRETATION

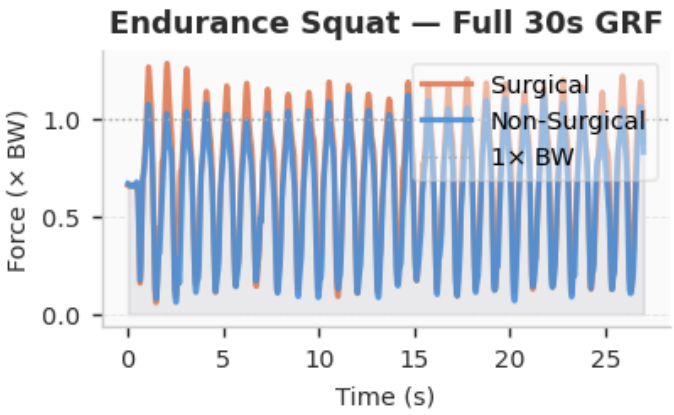
30s bilateral squat — reveals neuromuscular fatigue-induced asymmetry missed by single-rep testing.

Fatigue drift -8.3%: ⚠ LSI declined 8% — surgical limb offloads under prolonged effort. Endurance-specific rehab required.

Targets: Mean LSI ≥ 90% · Fatigue drift < 5% · No progressive offloading across 30s.

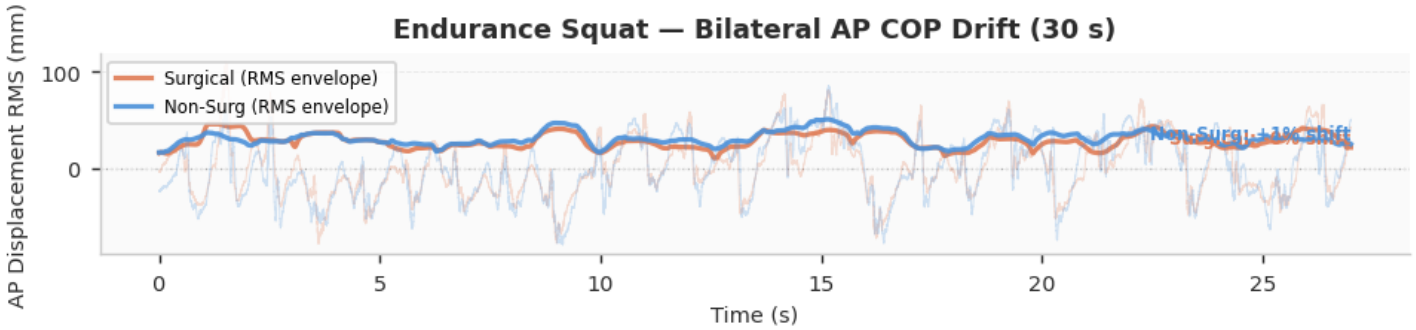


Per-rep LSI across 30s. Declining trend = neuromuscular fatigue breakdown.



Full 30s GRF trace. Force magnitude and rhythm across the set.

Left panel: *Surgical limb mean 30s force symmetry is symmetric with the non-surgical side (LSI 111%) — meets return-to-sport threshold.*



Bilateral AP COP drift across 30s (zero-centred, 1s RMS envelope). Surgical-side amplitude shrinking or trending posterior = offloading under fatigue. Diverging traces = progressive asymmetry with sustained effort.

Fatigue drift: -8.3% LSI change across 30s — surgical limb offloads progressively; 8% decline under fatigue.

Surgical limb mean force symmetry over 30 s is symmetric with the non-surgical side (LSI 111%) — meets return-to-sport threshold.

ABOUT THIS TEST

The 30-Second Endurance Squat assesses neuromuscular fatigue and limb symmetry under sustained bilateral loading. The athlete performs continuous bilateral squats to a standardised depth for a full 30 seconds. Unlike single-repetition tests, this task reveals how well the neuromuscular system maintains symmetry as it fatigues. Athletes post-ACL reconstruction frequently show an initial LSI within acceptable range that deteriorates progressively across the set as the nervous system disinhibits protective guarding. The COP drift trace exposes this compensation: surgical-limb weight transfer decreases, traces diverge, and loading shifts toward the uninvolved side. This test is particularly sensitive to residual quadriceps inhibition that is not apparent during brief, high-energy tasks.

SINGLE-LEG VERTICAL JUMP

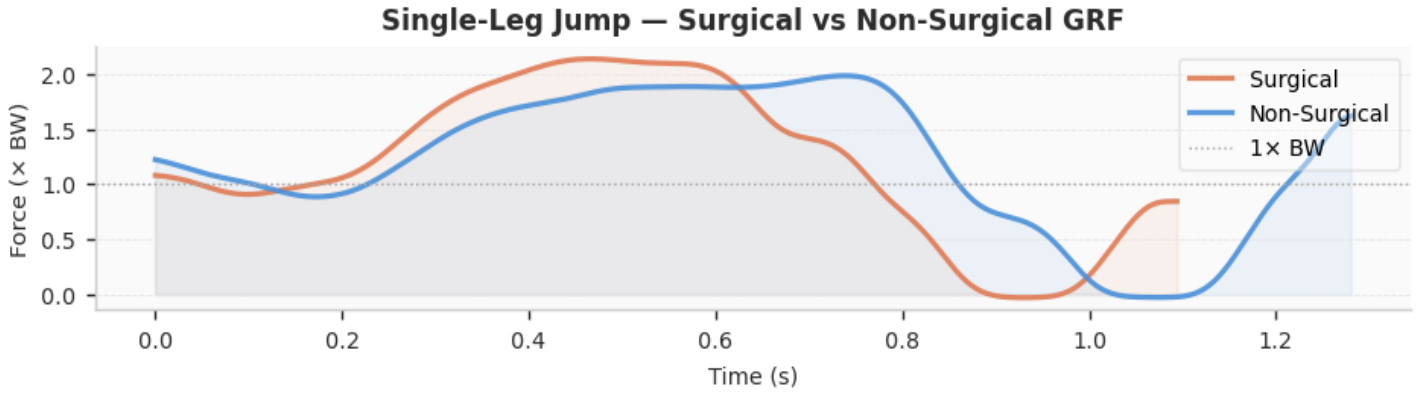
Metric	Surgical	Non-Surg	LSI / Δ
Jump Height (in)	7.1	5.9	119.2%
Peak GRF (N)	1805.1	1676.4	107.7%
Net Impulse (N·s)	309.2	316.1	97.8%
Contact Time (s)	0.89	1.02	—
Peak Knee Flexion (°)	62.5	60.6	—
Peak Knee Valgus (°)	8.4	-1.3	Δ +9.7°
Peak Tibial IR (°)	-10.3	-2.6	Δ -7.8°
Peak Hip Adduction (°)	12.3	4.2	Δ +8.0°
Peak Knee Ext Moment (Nm)	160.5	143.1	112.2%
AP COP Excursion (mm)	139.9	175.8	—

CLINICAL INTERPRETATION

Highest-demand test — isolates each limb, removing bilateral compensation. Strongest re-injury risk predictor.

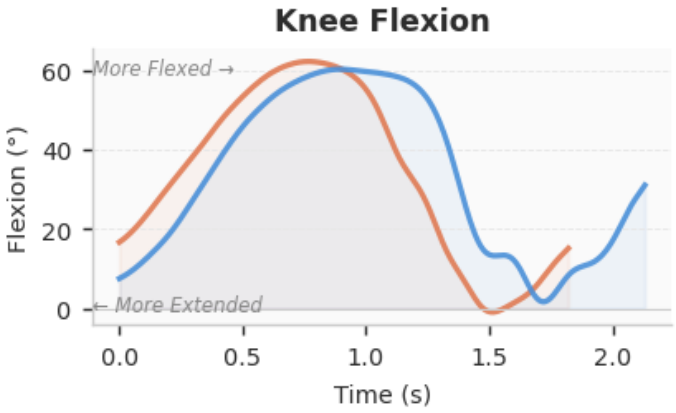
Jump height LSI 119%: ✓ Meets RTS threshold (≥ 90%).

Targets: Jump height LSI ≥ 90% · Peak GRF LSI ≥ 90% · Valgus < 8° · Tibial IR < 10° on surgical limb.

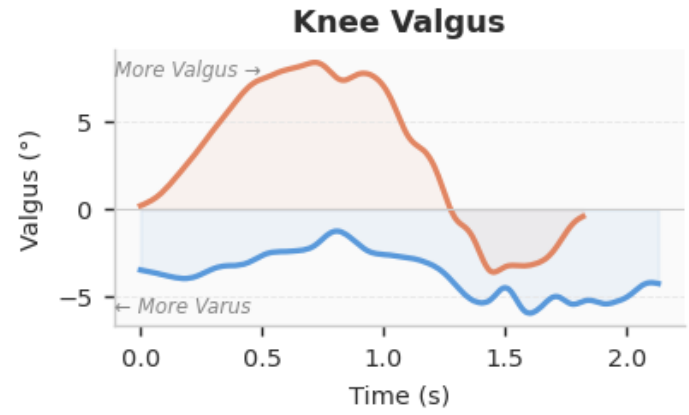


Per-limb GRF (separate trials). Propulsive peak height and shape should match.

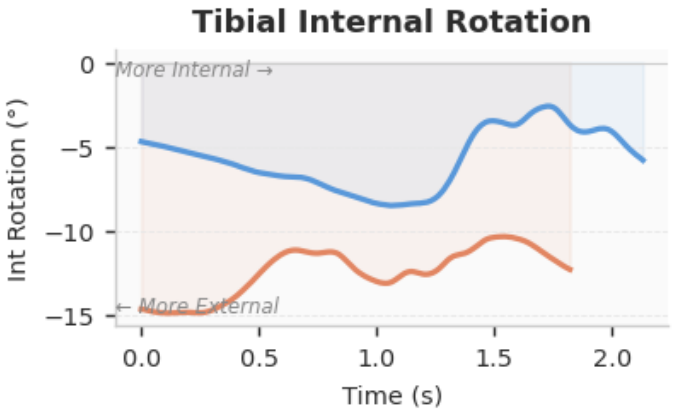
Surgical limb single-leg jump height is symmetric with the non-surgical side (LSI 119%) — meets return-to-sport threshold.



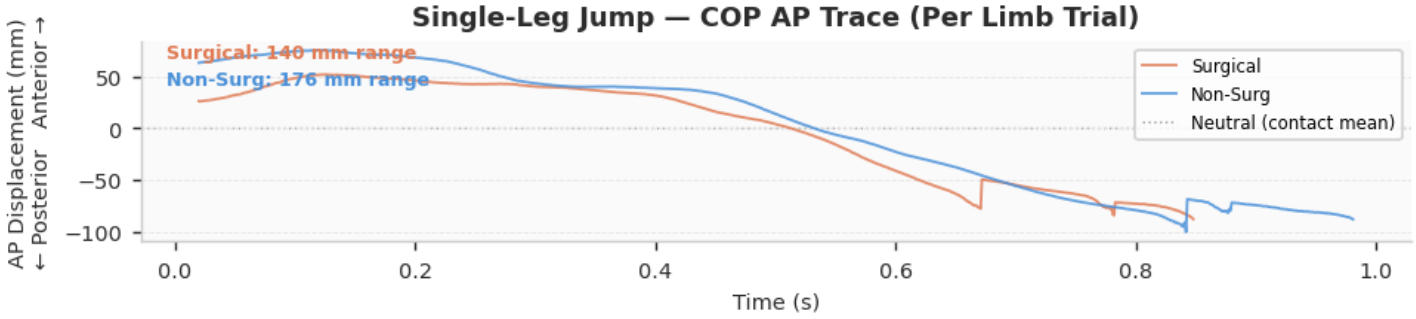
Knee flexion per limb. Reduced surgical depth = power deficit or guarded strategy.



Knee valgus per limb. Elevated surgical valgus under single-leg load = re-injury risk.



Tibial IR per limb. Surgical-side IR > 10° = high rotational ACL stress pattern.



AP COP during the take-off contact (zero-centred, per limb trial). Reduced anterior excursion on surgical side = guarded takeoff / limited push-off.

AP COP excursion: surgical 140 mm vs non-surgical 176 mm — less fore-aft travel on the surgical side — guarded or restricted loading.

Surgical limb single-leg jump height is symmetric with the non-surgical side (LSI 119%) — meets return-to-sport threshold.

ABOUT THIS TEST

The Single-Leg Vertical Jump isolates each limb independently, removing the ability of the uninvolved leg to compensate. The athlete performs a maximal countermovement jump landing on the same leg. Jump height LSI (surgical vs. non-surgical) is among the most sensitive power-based RTS criteria—a threshold of ≥ 90% is widely cited in ACL literature. Single-leg testing also reveals subtle compensations masked in bilateral tasks: reduced propulsive depth, hip drop (Trendelenburg pattern), and increased tibial rotation are common markers of residual neuromuscular asymmetry.

BALANCE — STANDARD (EYES CLOSED, FIRM SURFACE)

Metric	Surgical	Non-Surg	LSI / Δ
COP Mean Velocity (mm/s)	157.7	100.2	63.6%
COP Ellipse Area (mm²)	14078.4	4319.7	30.7%
Total Excursion (mm)	2207.3	1403.2	—
AP Range (mm)	106.0	89.7	—
ML Range (mm)	278.9	53.1	—
RMS Displacement (mm)	45.7	22.2	—

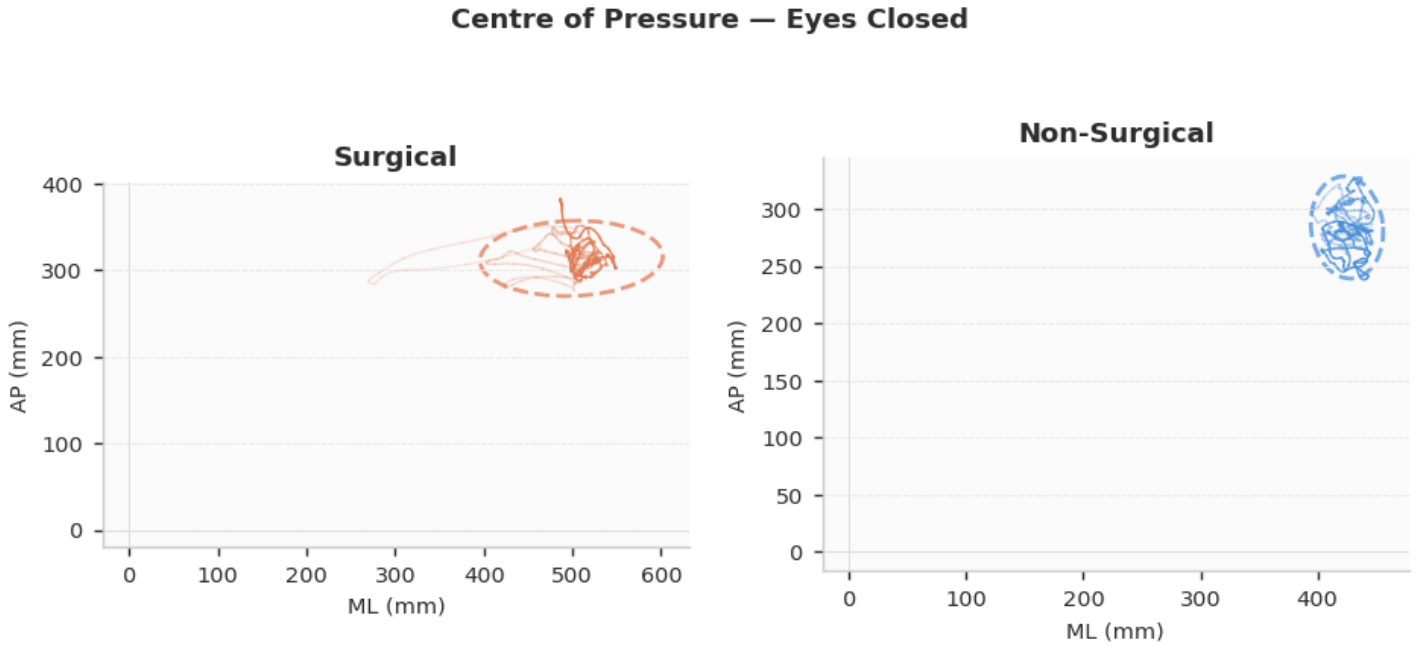
CLINICAL INTERPRETATION

Eyes-closed single-leg stance on firm surface. COP velocity is the most sensitive metric — higher speed = more corrective activity = reduced proprioception.

Surgical 158 mm/s vs Non-Surgical 100 mm/s — Surgical limb less stable. Proprioceptive retraining indicated.

COP Velocity LSI 64% (NS÷Surg×100; below 100% = surgical side sways more). LSI target ≥ 90%. COP low-pass filtered at 10 Hz.

⚠ **Data check:** COP range on the surgical trial exceeds 150 mm — likely a foot shift or touchdown rather than sway. Review / re-collect before relying on this page.



COP trajectory (10 Hz low-pass) — eyes closed. Tighter ellipse = steadier stance. 95% confidence ellipse shown.

ABOUT THIS TEST

The Balance / Proprioception test assesses joint position sense and neuromuscular control under sensory challenge. The athlete stands on a single leg with eyes closed, removing visual compensation and isolating the mechanoreceptor feedback system of the ankle and knee. Center of Pressure (COP) velocity is the most sensitive metric: a higher velocity indicates greater corrective activity, meaning the neuromuscular system is working harder to maintain stability. After ACL reconstruction, deficits in mechanoreceptor density within the graft and capsular tissue result in elevated COP velocity and a larger sway ellipse on the surgical limb. The Airex pad condition adds an unstable surface to further stress the proprioceptive system, revealing deficits that may be masked on firm ground. Symmetry between limbs (LSI ≥ 85–90%) is the primary RTS criterion for this test.

BALANCE — AIREX PAD (EYES CLOSED, UNSTABLE SURFACE)

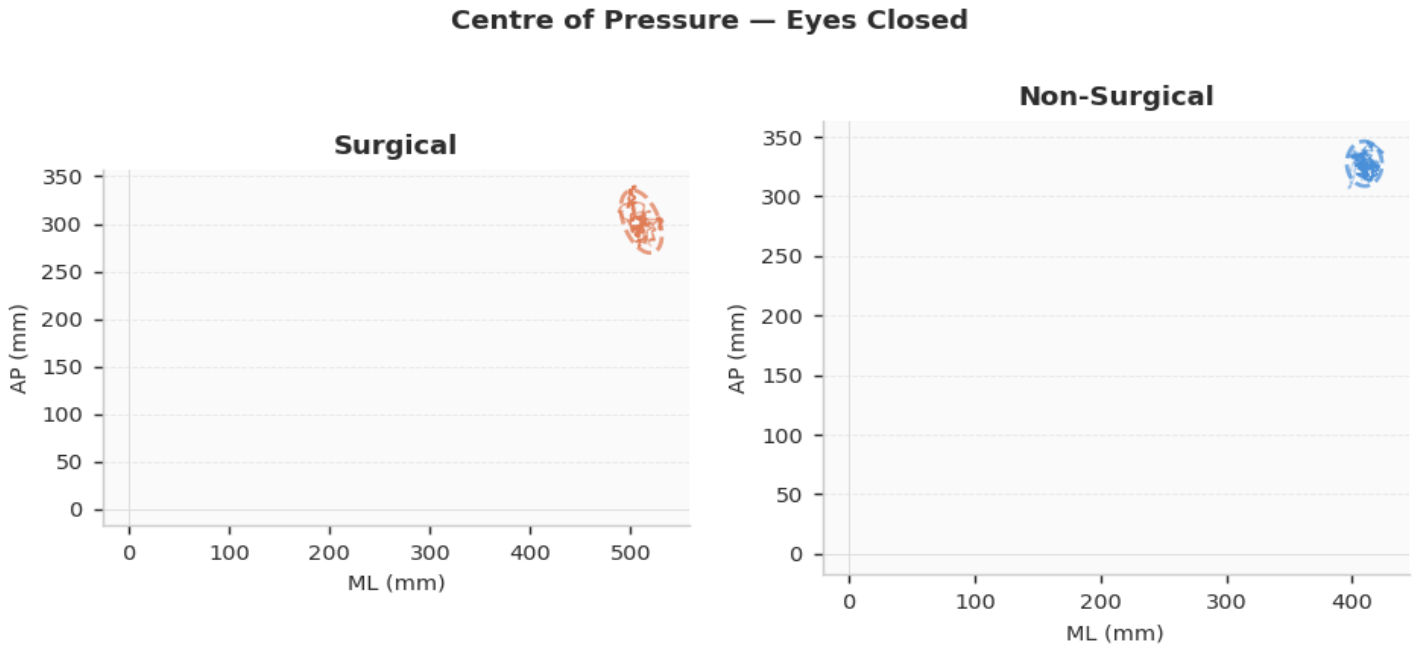
Metric	Surgical	Non-Surg	LSI / Δ
COP Mean Velocity (mm/s)	55.5	55.2	99.6%
COP Ellipse Area (mm²)	1904.4	821.5	43.1%
Total Excursion (mm)	776.6	773.1	—
AP Range (mm)	63.3	39.0	—
ML Range (mm)	43.9	27.9	—
RMS Displacement (mm)	15.9	9.5	—

CLINICAL INTERPRETATION

Eyes-closed single-leg stance on Airex foam pad. Higher COP values are expected — the unstable surface challenges proprioceptive and neuromuscular systems. Symmetry between limbs (LSI ≥ 85%) is the key target.

Surgical 55 mm/s vs Non-Surgical 55 mm/s — COP velocity comparable between limbs.

COP Velocity LSI 100% (NS÷Surg×100; below 100% = surgical side sways more). LSI target ≥ 90%. COP low-pass filtered at 10 Hz.



COP trajectory (10 Hz low-pass) — eyes closed. Tighter ellipse = steadier stance. 95% confidence ellipse shown.

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